

Problem 1a

`brown_cfd[w]` returns a `FreqDist` that counts the words that follow `w` in the Brown text.

`brown_cfd[w1][w2]` is the amount of times that `w2` follows `w1` in the Brown text.

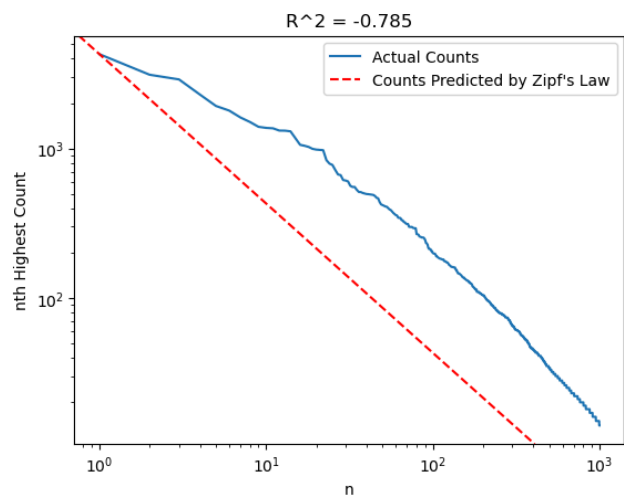
Problem 1b

`brown_char_cfd` is a `Conditional FreqDist` that, instead of looking at the word-level like `brown_cfd`, looks at the character level.

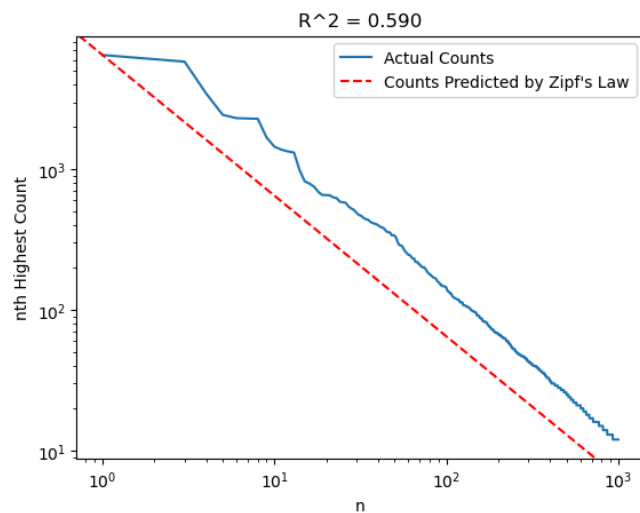
For each key in `brown_char_cfd`, only ever a character, it returns the distribution, in quantity, of the next character as present in the Brown text. `brown_cfd` does this for words.

Problem 2a

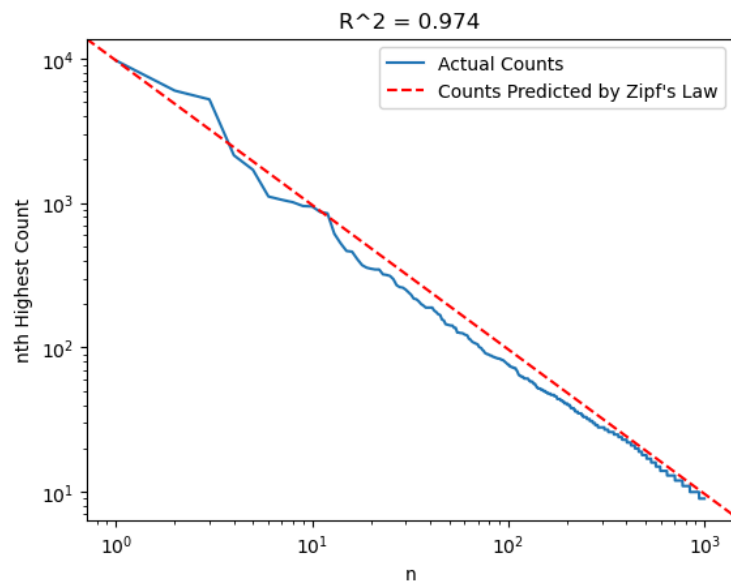
Voynich



Orwell English



Orwell Hungarian



English and the Voynich manuscript hover above the zipfian line, where hungarian matches it more consistently. The Voynich manuscript even has a negative R^2 .

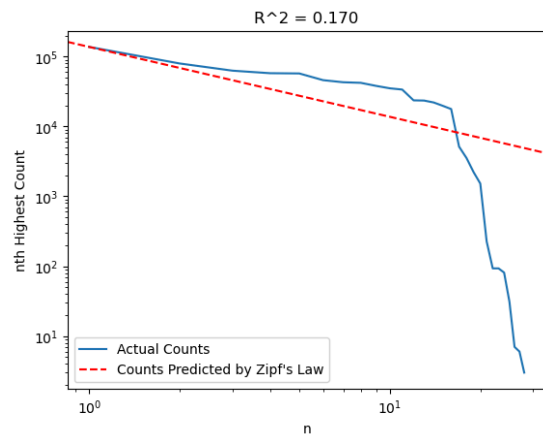
Problem 2b.

Corpus	#of Repetitions	Length	% Repititions
Brown	6,166	1,161,192	.531%
Voynich	6,444	137,320	4.69%
1984(en)	16	118,424	.0135%
1984(hu)	27	98,426	.0274%

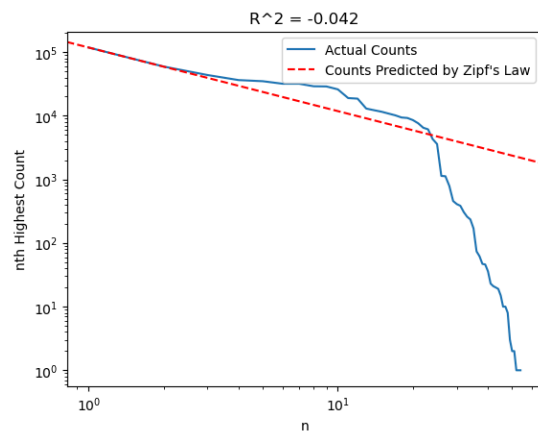
This could result from Orwell's writing style, or from spacing in the top two corpora.

Problem 2c

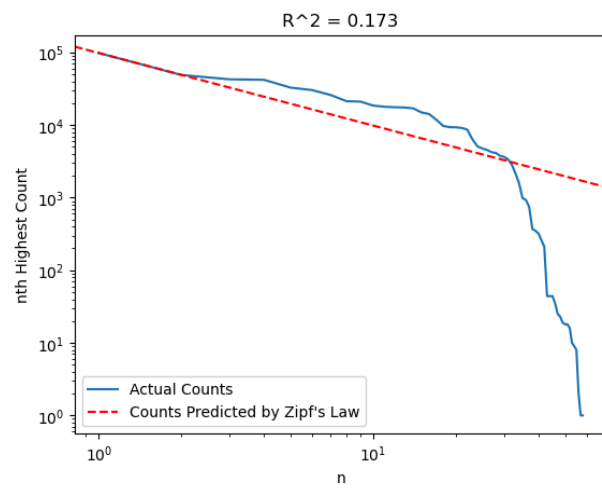
Voynich



Orwell English



Orwell Hungarian



In general character frequencies are NOT zipfian

Problem 2d

Corpus	Conditional Character Entropy
Brown	3.41
Voynich	2.21
1984(en)	3.31
1984 (hu)	3.58

Problem 2e

Brown	Voynich	1984(en)	1984(hu)
qu: .981	qo: .974	âc: .1.0	wi: .989
ve: .651	ch: .822	qu: .1.0	cs: .587
he: .479	sh: .634	ve: .729	ül: 0.575
ze: .444	dy: .536	ze: .578	ít: 0.471
ke: .335	ai: .460	ju: .547	xu: 0.44
ju: .317	he: .454	he: .470	gy: 0.377
be: .308	pc: .432	be: .342	fe: 0.368
th: .307	in: .430	th: .329	ha: 0.356
jo: .281	ii: .412	ke: .308	va: 0.292
in: .259	fc: .364	in: .305	va: 0.292

Problem 2f

No, the most predictable character-level bigrams are not digraphs. Of the top ten digraphs in English, âc, present once in the corpus, is a result of the singular appearance of papier mâché, and does not indicate digraph-ness, given the relegation of digraphs to few loan words in English. ‘qu’ is indeed a digraph, but only one of the remaining next 8 is a digraph, ‘th’ representing the dental fricative.

Hungarian is similarly representative of digraphs, with only cs and gy being digraphs.

I speculate that this disappointing answer comes from the commonality of the individual graphs that compose the digraphs, or in similar digraphs splitting the vote. In English, t is the second most common letter in the alphabet. I posit instead that the most common bigrams instead originate from the most common words, or phonotactical or historical rules governing the orthography (as how v often appears at the end of words with a silent e)